



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
- Discussion
- Conclusion



Executive Summary

- **Summary of methodologies**

- In Applied Data Science Capstone project, we will try to predict if the SpaceX's Falcon 9 first stage will land successfully, using more than one Machine learning algorithms.
- Different stages of the Capstone project:
 - Data collection, wrangling and formatting
 - Exploratory Data Analysis
 - Interactive Visual Data Analytics
 - Machine Learning Predictions

- **Summary of all results**

- Success/Failed launches were closely related to some of the rocket launches characteristics from the graphs' outcomes. KSC LC-39A had the highest success rate of all launch sites
- ES-L1, SSO, HEO and GEO had the highest success rates among the Orbits.
- It was determined that the best machine learning model with best accuracy to predict success/failed launches for Falcon 9 first stage could be the DecisionTreeClassifier.

Introduction

- **Project background and context**

- In this capstone, we will predict if
 - the Falcon 9 first stage will land successfully.
- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars;
- other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage.
- Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.
- Most unsuccessful landings are planned. Space X performs a controlled landing in the oceans.

- **Problems you want to find answers**

- We are trying to find out if using a set of Falcon 9 rocket launches features such as launch site , orbit type, payload mass and so on will the Falcon 9 rocket land successfully or not?

Section 1

Methodology

Methodology

Executive Summary

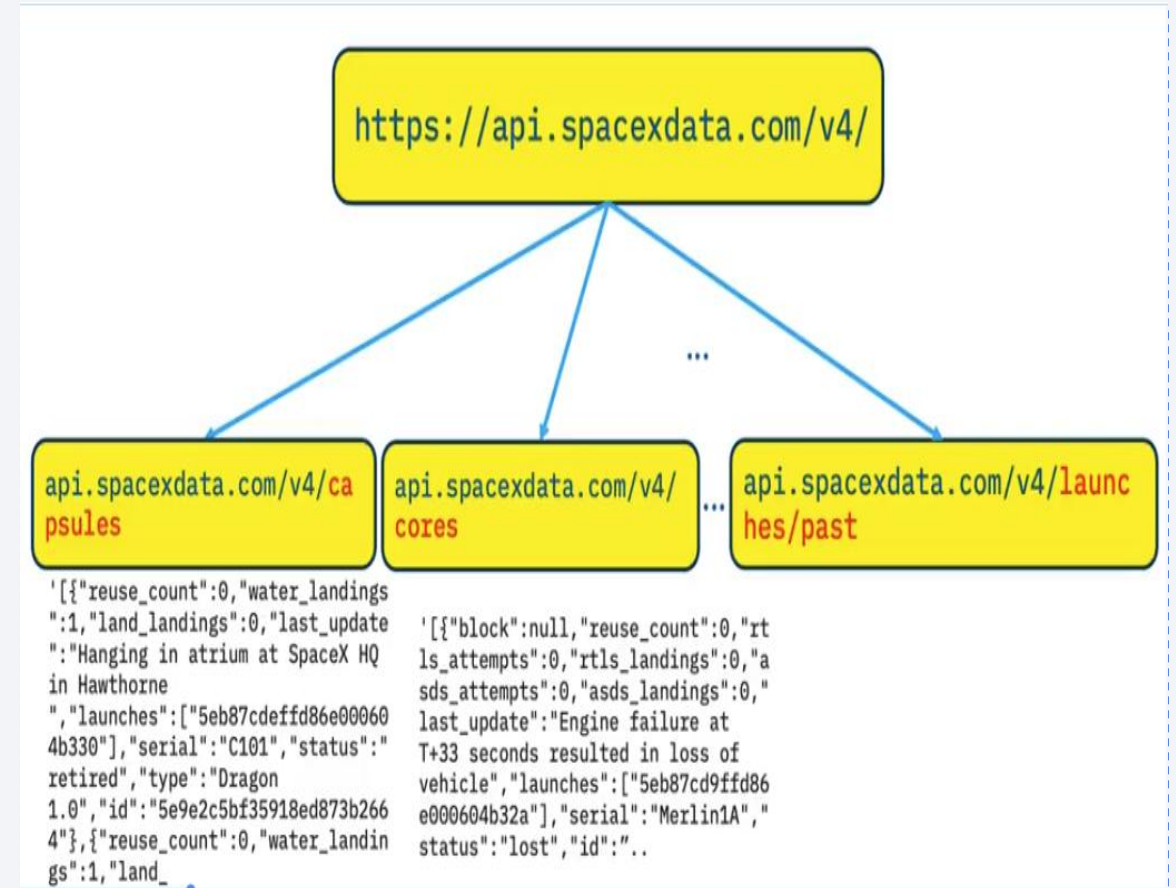
- Data collection methodology:
 - **SpaceX API**
Using `get(https://api.spacexdata.com/v4/rockets/)`
 - **Web scraping to collect Falcon 9 historical launch records from:**
 - https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
 - **Data wrangling**
 - Find some patterns in the data
 - Determine what would be the label for training supervised models.
 - **Exploratory data analysis (EDA) using SQL and visualization**
 - Visualize the relationship between features
 - Feature engineering
 - **Interactive visual analytics using Folium and Plotly Dash**
 - Mark all launch sites on a map
 - Mark the success/failed launches for each site on the map
 - Calculate the distances between a launch site to its proximities
 - Build an Interactive Dashboard of the data
 - **Predictive analysis using classification models**
 - Find the method performs best using several machine learning classification algorithms

Data Collection – SpaceX API

- Collected data and verified that is in the correct format using SpaceX API request.
 - requests.get(<https://api.spacexdata.com/v4/rockets/>)
 - requests.get(<https://api.spacexdata.com/v4/launchpads/>)
 - requests.get(<https://api.spacexdata.com/v4/payloads/>)
 - requests.get(<https://api.spacexdata.com/v4/cores/>)
- **From cores we would like to learn the outcome of the landing, the type of the landing, number of flights with that core, whether gridfins were used, whether the core is reused, whether legs were used, the landing pad used, the block of the core which is a number used to separate version of cores, the number of times this specific core has been reused, and the serial of the**

Data Collection – SpaceX API

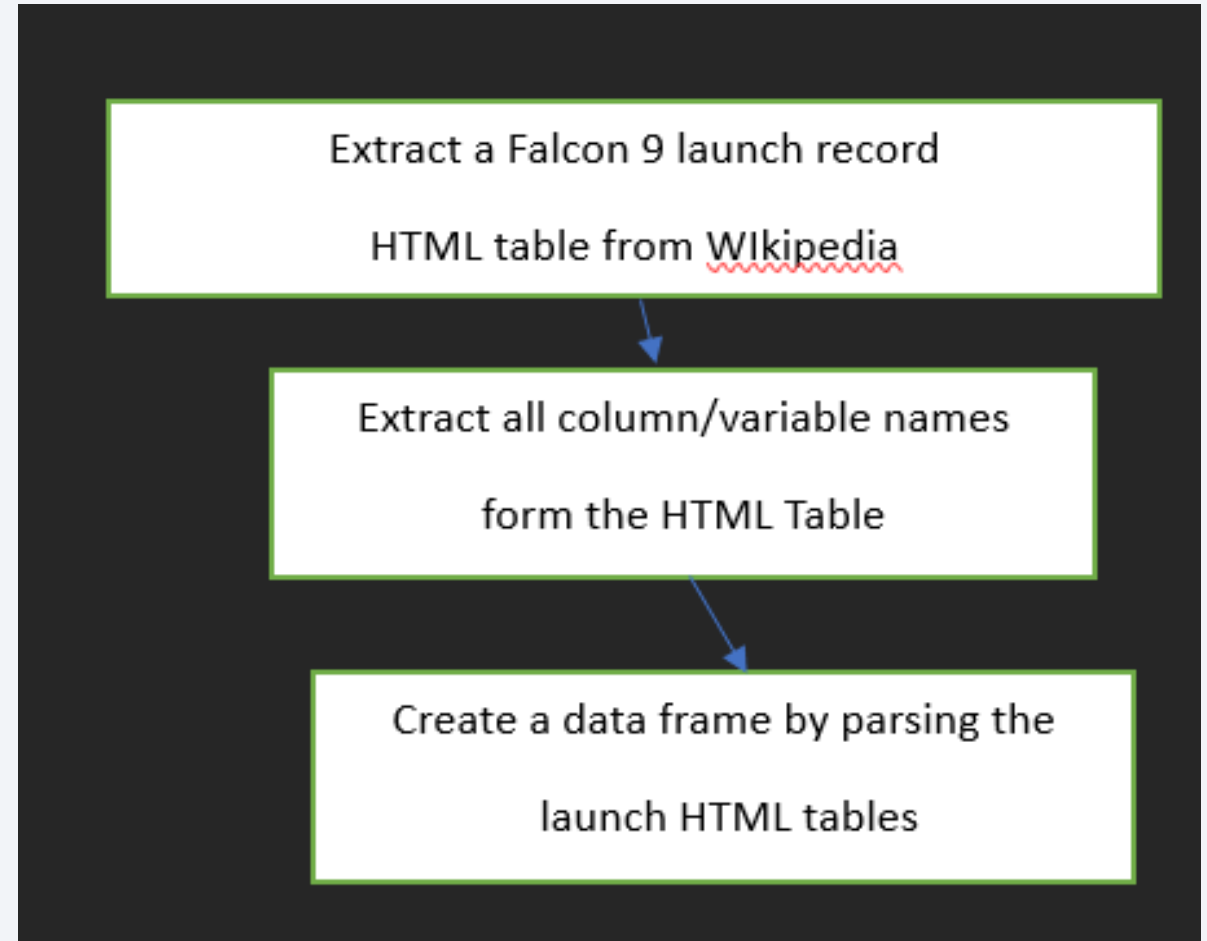
- The SpaceX REST API endpoints, or URL, starts with `api.spacexdata.com/v4/`. We have the different end points, for example: `/capsules` and `/cores`. We will be working with the endpoint `api.spacexdata.com/v4/launches/past`. Let's see how the API works.
- Notebook (GitHub URL)
[Github URL](#)



Data Collection - Scraping

- Extract a Falcon 9 launch record
HTML table from Wikipedia
- Process web scraped
HTML tables using
BeautifulSoup to extract the
tables launch data
- Create a data frame by parsing
the launch HTML tables
- Notebook (GitHub URL)

Github URL



Data Collection – Wrangling

- Perform some Exploratory Data Analysis (EDA) to find some patterns in the data
- Determine The label for training supervised models.
- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Create a landing outcome label from Outcome column
 - Outcome class = 1 success and class = 0 failed
- Notebook (Github URL)

Github URL

EDA with Data Visualization

- Visualize the relationship between Payload Mass and Launch Site or any other variable we understand the variable's importance.
 - Payload Mass and Launch Site
 - Orbit and Outcome
 - Payload and Orbit
 - Yearly success rate
 - Flight number and Orbit
 - Flight Number and Launch Site
- Create dummy variables to categorical columns
- Cast all numeric columns to float64
- Notebook (Github URL):
Github URL

EDA with SQL

- Summary of the SQL queries performed
 - Display the names of the unique launch sites in the space mission
 - Display 5 records where launch sites begin with the string 'CCA'
 - Display the total payload mass carried by boosters launched by NASA (CRS)
 - Display average payload mass carried by booster version F9 v1.1
 - List the date when the first successful landing outcome in ground pad was achieved.
 - boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - List the total number of successful and failure mission outcomes
 - Booster versions that have carried the maximum payload mass,
 - booster versions, launch site for the months in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) between the date 2010-06-04 and 2017-03-20, in descending order.

Notebook (Github URL)

Github URL

Build an Interactive Map with Folium

- Analyze Launch Site Geo Data with Folium map
 - Mark the locations and proximities of launch sites
 - Discover patterns via exploring the map faster and more effectively
 - Explain how to choose an optimal launch site
 - More appalling stories
 - Explain why you added those objects
- Notebook (GitHub URL)

Github URL

Build a Dashboard with Plotly Dash

- An interactive dashboard application contains input components such as a dropdown list and a range slider to interact with a pie chart and a scatter point chart was build using Plotly Dash
- Adding those plots and interactions help us to find more insights from the SpaceX dataset more easily than with static graphs.
- Plotly Dash App (GitHub URL)

Github URL

Predictive Analysis (Classification)

- Build a machine learning pipeline to predict if the first stage of the Falcon 9 lands successfully.
- This will include:
 - Standardize our data, and Train test split,
 - Split our data into training and testing data
 - Use GridSearchCV, to test hyperparameters for several algorithm:
 - Logistic Regression
 - SVC
 - Decision Tree Classifier
 - k Neighbors Classifier
- GitHub URL:
[Github URL](#)

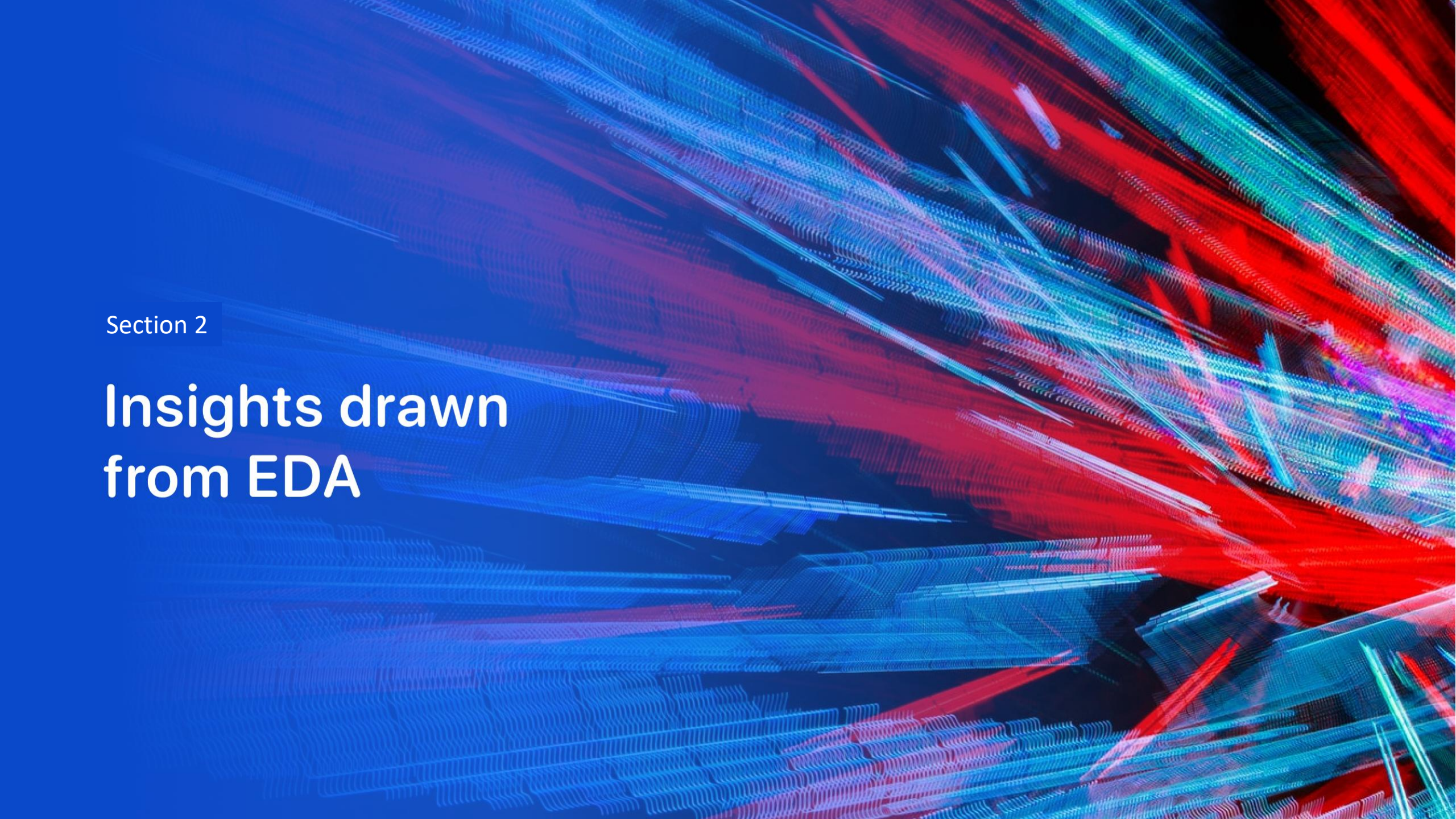
Build a Machine Learning Pipeline

- Predict whether first stage of Falcon 9 will land successfully



Results

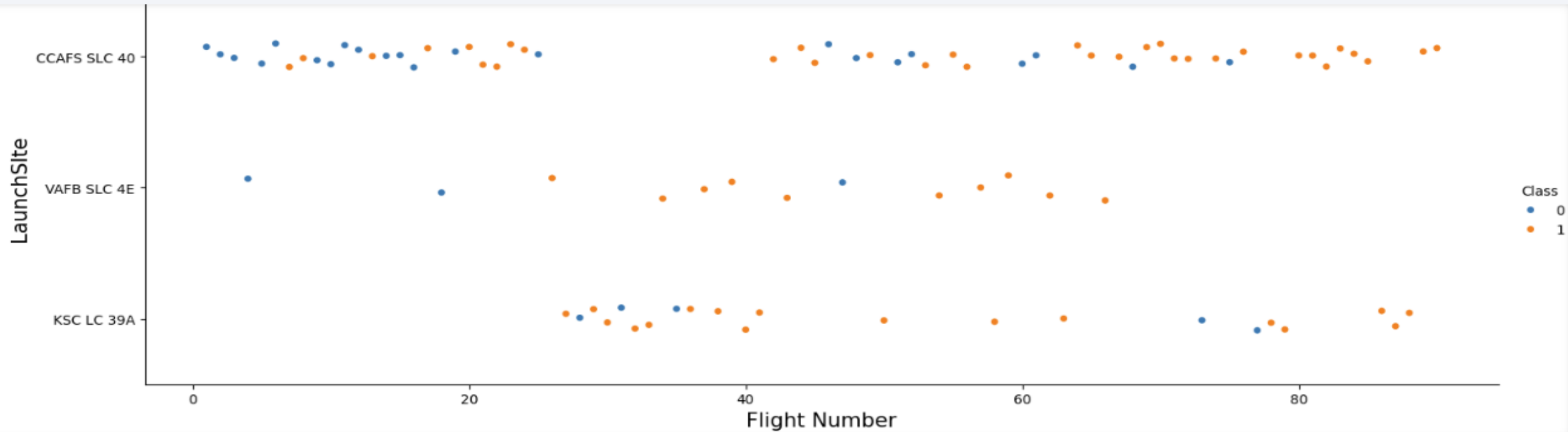
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue gradient on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and teal on the right. These streaks vary in thickness and intensity, creating a sense of motion and depth. A faint, light blue grid pattern is visible across the entire background, adding a technical or digital feel to the design.

Section 2

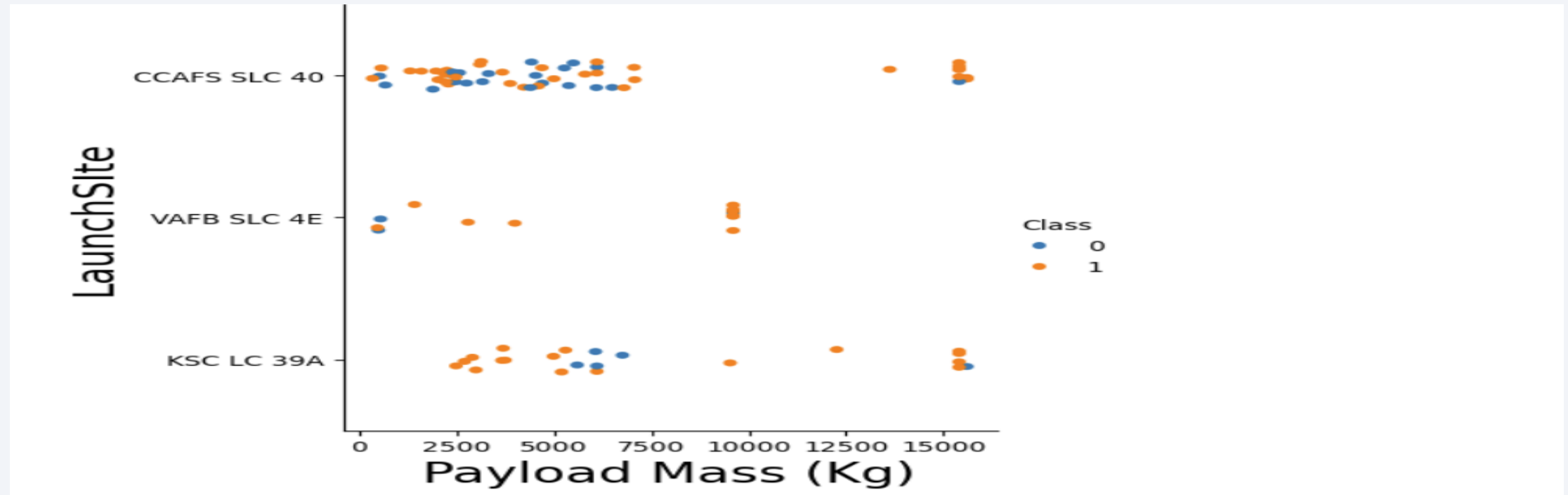
Insights drawn from EDA

Flight Number vs. Launch Site



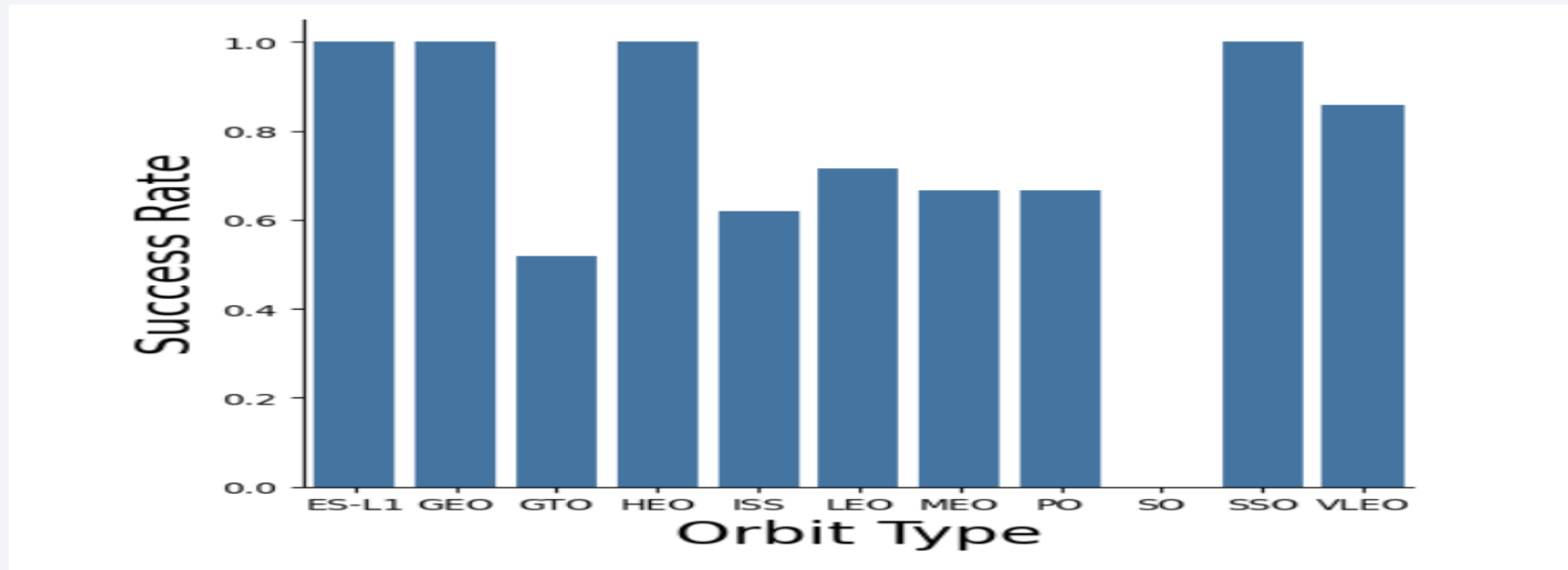
- All sites shows familiar and success. As the number of fights increases the success rate gets better
- Most landing flights on CCAFS SLC-40 site and most successful landing flights on VAFB SLC 4E.
- Class = 0 (blue color) failed landing and Class = 1 (red color) successful landing

Payload vs. Launch Site



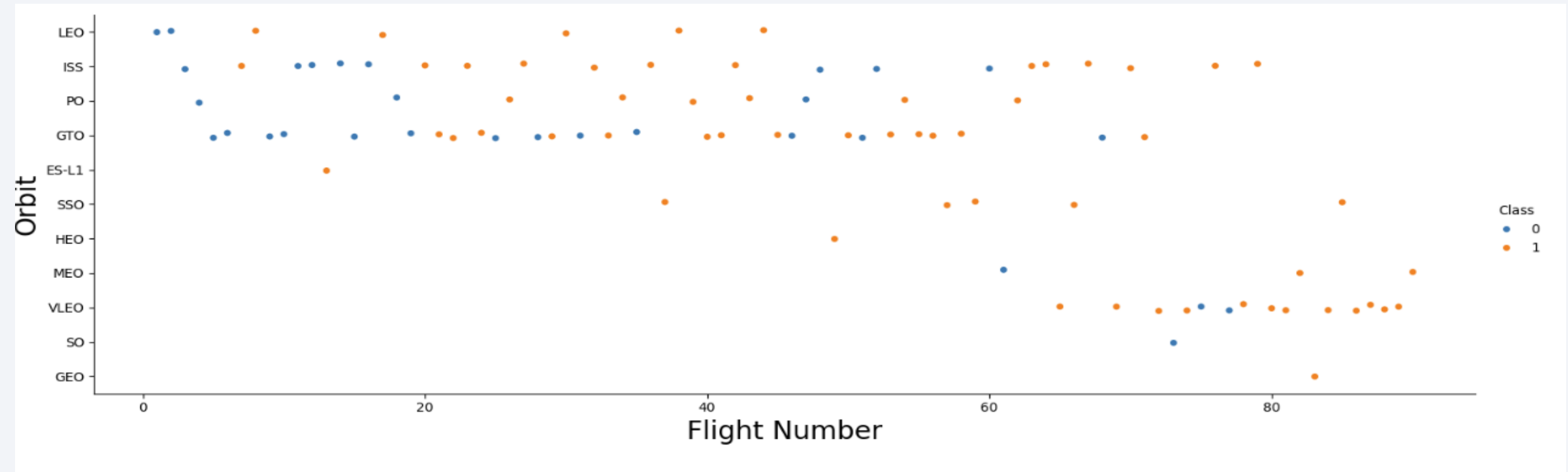
- The lighter the payload the higher landing failures at early flights
- As the payload increase the higher successful landing outcomes
- KSC LC 39A site has the most successful landing flights

Success Rate vs. Orbit Type



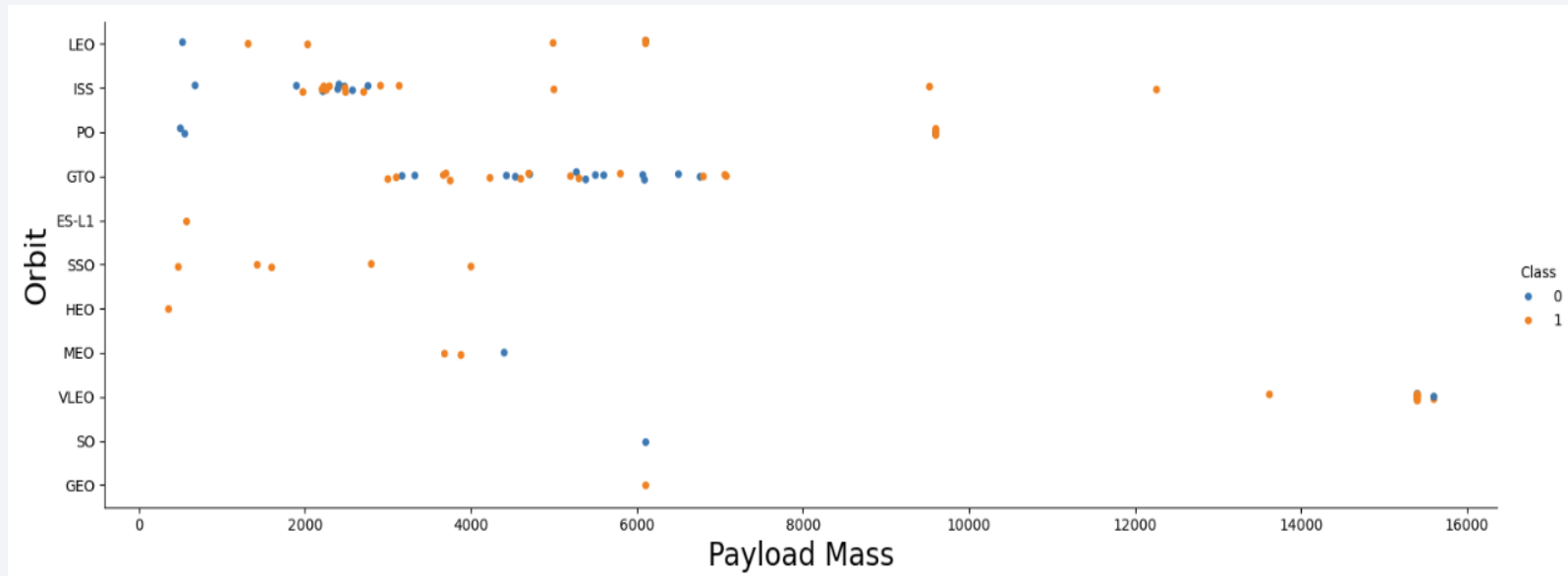
- ES-L1, GEO, HEO, and SSO orbits show the highest success rates.
- GTO orbit shows the lower success rate
- SO orbit type does not have enough data to show any accurate analysis

Flight Number vs. Orbit Type



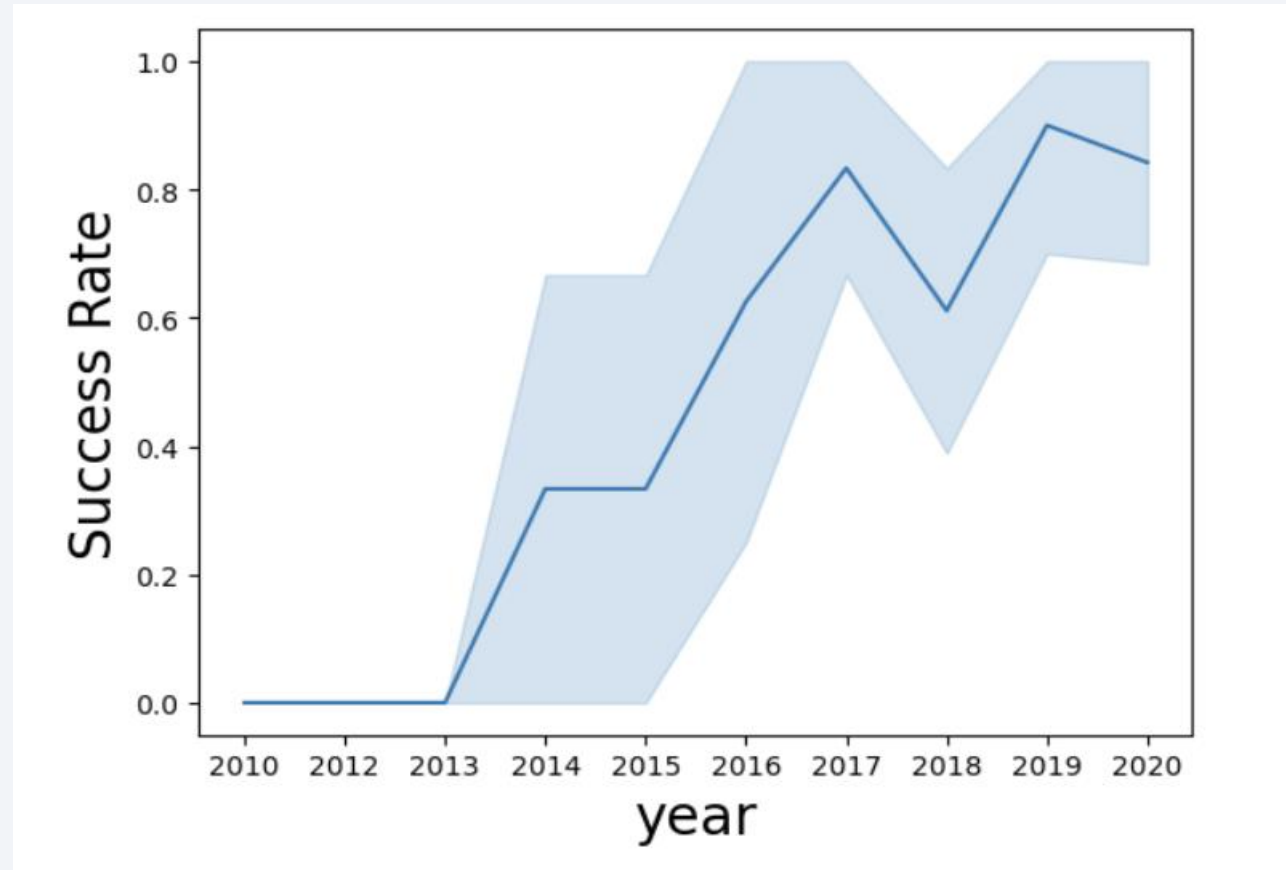
- A number of orbits show no flights at the beginning. It could be a sign of waiting for improvement as flight numbers increase.
- Flight numbers increase success rate increase as well. More experience and improvements.

Payload vs. Orbit Type



- A number of orbits show across a big range of payload mass.
- HEO, MEO, GEO and LEO show a lower range and high rate of successful landing
- PO and LEO show successful landing with higher payload mass
- Hard to distinguish the relationship between Orbit type and Payload mass both success/fail is present

Launch Success Yearly Trend



- Show success rate keep increasing since 2013 to 2020

All Launch Site Names

- The unique launch sites are:
 - CCFS LC-40
 - CCAFS SLC-40
 - KSC LC-39A
 - VAFB SLC-4E
- #SELECT DISTINCT "Launch_Site"
- #FROM SpaceX";
- %sql SELECT DISTINCT(LAUNCH_SITE FROM SPACEXTBL;
- * sqlite:///my_data1.db

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- `%sql SELECT * FROM SPACEXTBL WHERE (LAUNCH_SITE ) LIKE '%CCA%' Limit 5;`

Total Payload Mass

- The total payload carried by boosters from NASA (CRS):
 - **45,596kg**
- Query:
 - **%sql SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';**
 - Use the SpaceX table to extract the payload mass kg for NASA (CRS)

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is:
 - 2,928.4kg
- %sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTBLE WHERE Booster_Version = 'F9 v1.1',
- Use SpaceX table to extract the avg payload mass kg for booster version
 - F9 v1.1

First Successful Ground Landing Date

- The first successful landing outcome on ground pad was achieved

2015-12-22

- Query :
- **%sql SELECT min(Date) FROM SPACEXTBL WHERE**
- “Landing_Outcome” = ‘Success (ground pad)’ :
- * [sqlite:///my_data1.db](#)

Successful Drone Ship Landing with Payload between 4000 and 6000

- The boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- Query:

- **%sql SELECT PAYLOAD FROM SPACEXTB WHERE**

“Landing_Outcome” = ‘Success(drone Ship)’ \ AND

4000 < “PAYLOAD MASS_KG “ < 6000;

*** [sqlite:///my_data1.db](#)**

Payload
SpaceX CRS-8
JCSAT-14
Thaicom 8
JCSAT-16
Iridium NEXT 1
SES-10
BulgariaSat-1
Iridium NEXT 2
Formosat-5
Iridium NEXT 3
SES-11 / EchoStar 105
Koreasat 5A
Transiting Exoplanet Survey Satellite (TESS)
Bangabandhu-1

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes

Mission Outcome	Total number
Success	100
Failure	1

- Query:
- `%sql SELECT Mission_Outcome, COUNT(*) AS "total_number" FROM SPACEXTBL GROUP BY Mission_Outcome;`
- * sqlite:///my_data1.db

Boosters Carried Maximum Payload

- The boosters which have carried the maximum payload mass ->

- Query that list all the booster versions that have carried the maximum payload mass:

```
%sql SELECT DISTINCT BOOSTER_VERSION FROM SPACEXTBL  
WHERE PAYLOAD_MASS_KG = (SELECT MAX(PAYLOAD_MASS_KG)  
FROM SPACEX);
```

* sqlite:///my_data1.db

Booster Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List of failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

Landing Outcome	Date	Booster Version	Launch Site
Failure (drone ship)	1/10/2015	F9 v1.1 B1012	CCAFS LC-40
Failure (drone ship)	4/14/2015	F9 v1.1 B1015	CCAFS LC-40

```
%sql SELECT "LANDING_OUTCOME", DATE, BOOSTER_VERSION, LAUNCH_SITE FROM SPACEXTBL \
WHERE "Landing_Outcome" = 'Failure (drone ship)' AND substr(Date,0,5) = '2015';
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Query:

Landing Outcome	total number
Success	38
No attempt	21
Success (drone ship)	14
Success (ground pad)	9
Failure (drone ship)	5
Controlled (ocean)	5
Failure	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1
No attempt	1

```
%sql SELECT "LANDING_OUTCOME", COUNT(LANDING_OUTCOME) AS total_number FROM SPACEXTBL \
WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "LANDING_OUTCOME" ORDER BY total_number DESC;
```

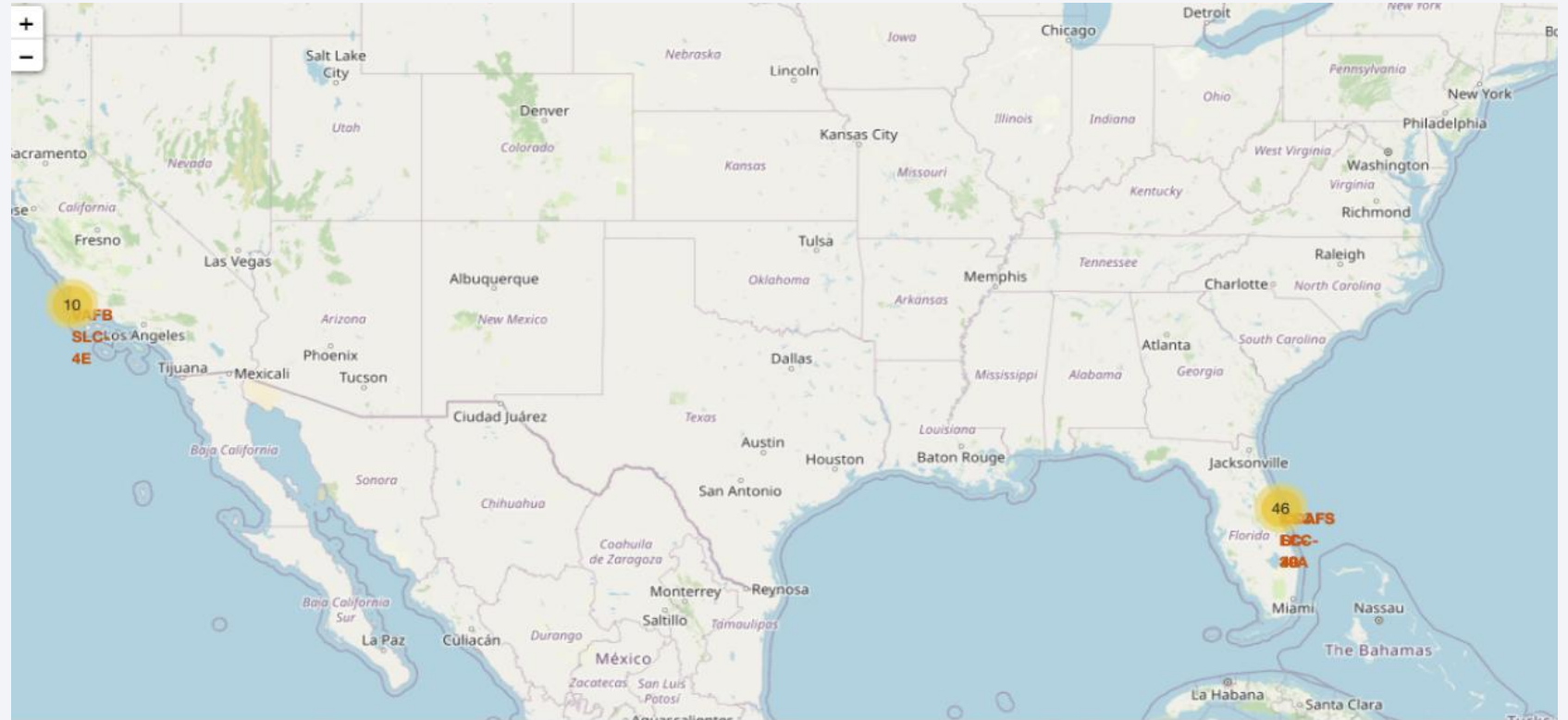
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

launch Sites' Location

- Launch sites are
Located near
California
and Florida
coastal line
incase of a failure
avoiding any human
injuries
- Launch sites are seen
In yellow circles on the map



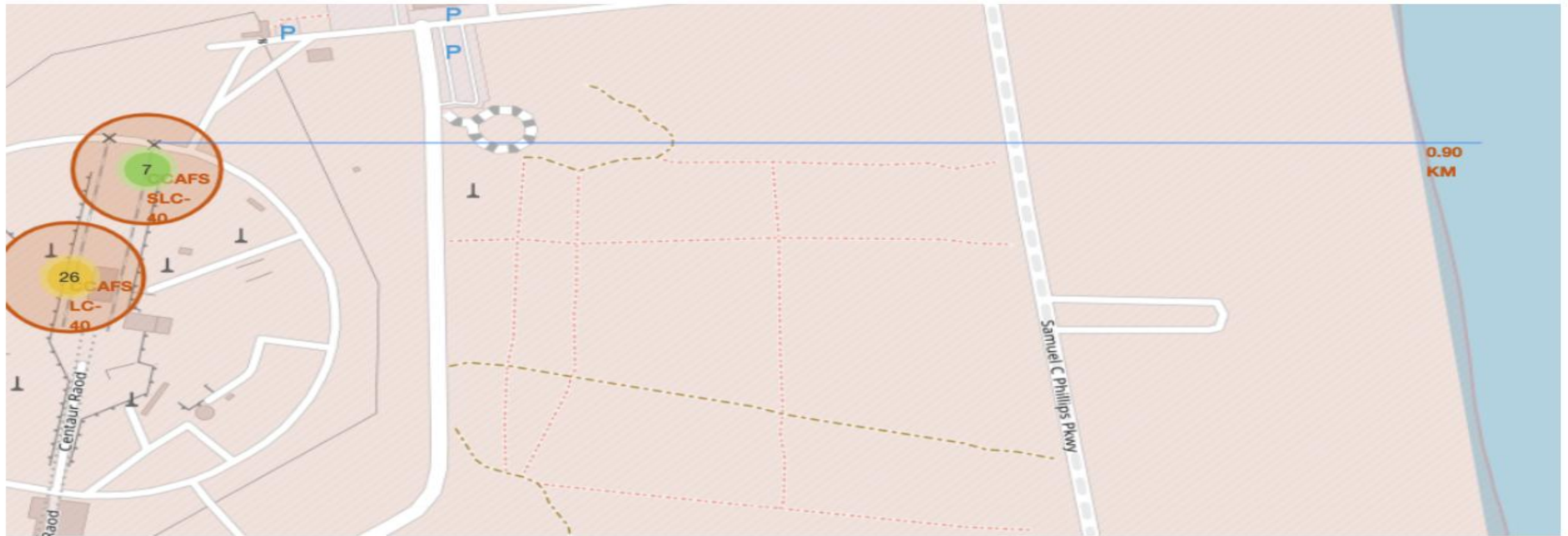
Launch Outcomes on the map

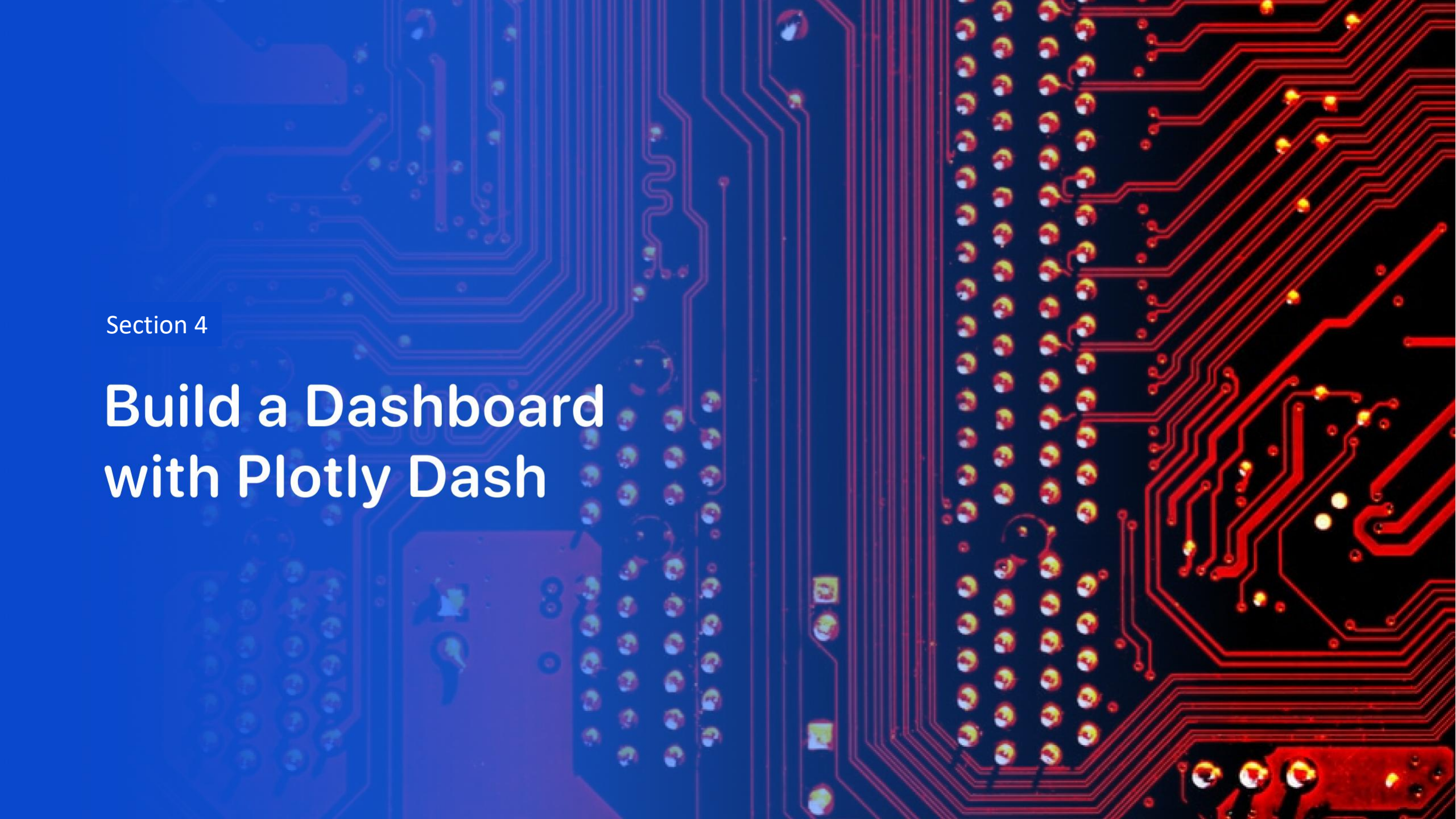
- If the launch was successful was marked Green (class = 1)
If the launch failed, it was marked red (class = 0)



Launch site Proximities

- Distance to Coastline (.90KM)





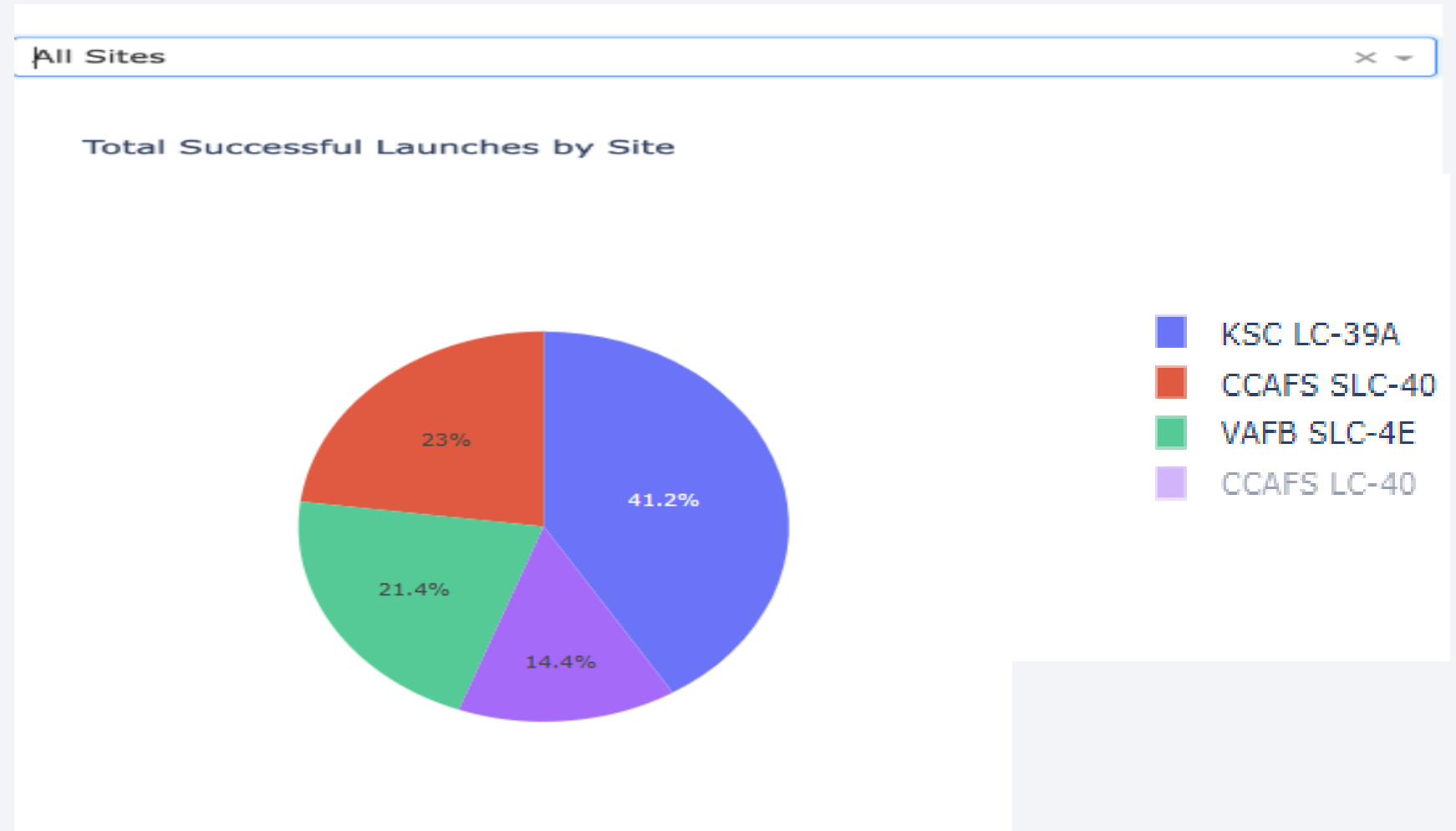
Section 4

Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

KSC LC-39A had the highest ratio of successful landing, followed by CCAFS SLS-40

VAFB SLC-4E and CCAFS LC-40 the lowest

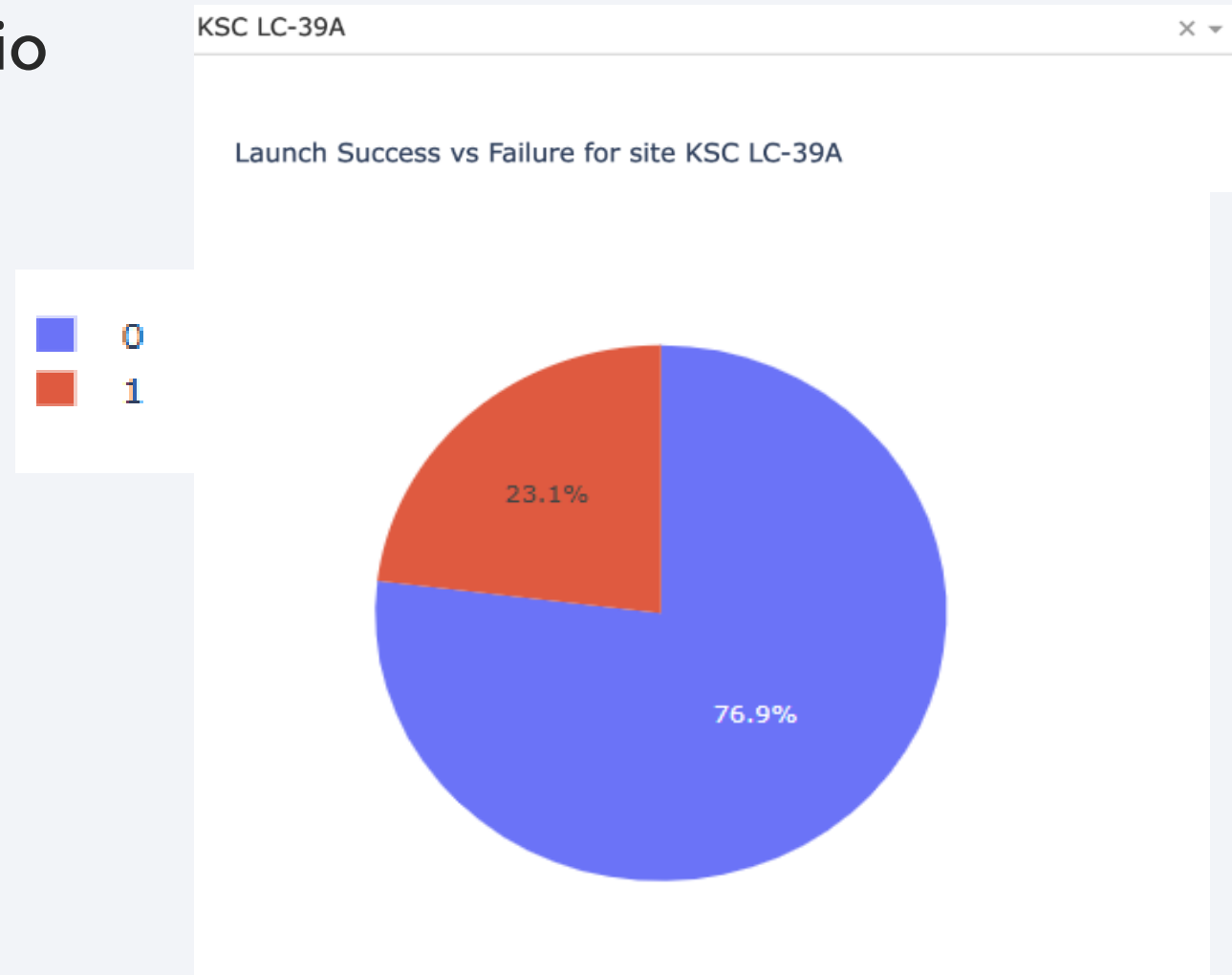


All Sites Success Ratio - Highest

- **KSC LC -39A** had the highest ratio of successful landings

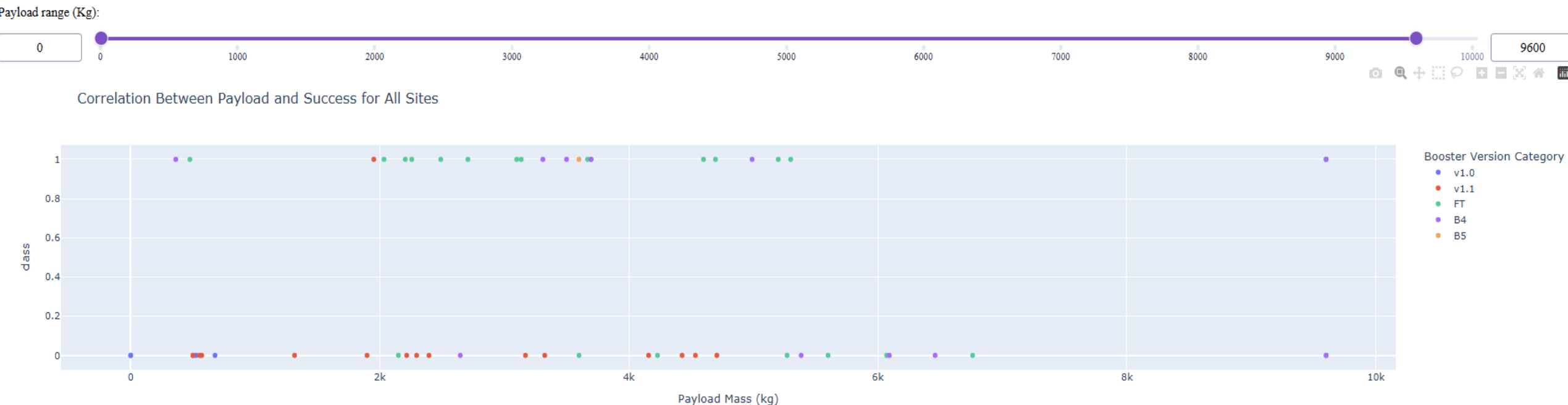
1 represent success

0 represent failed



SpaceX Dash – Payload Range

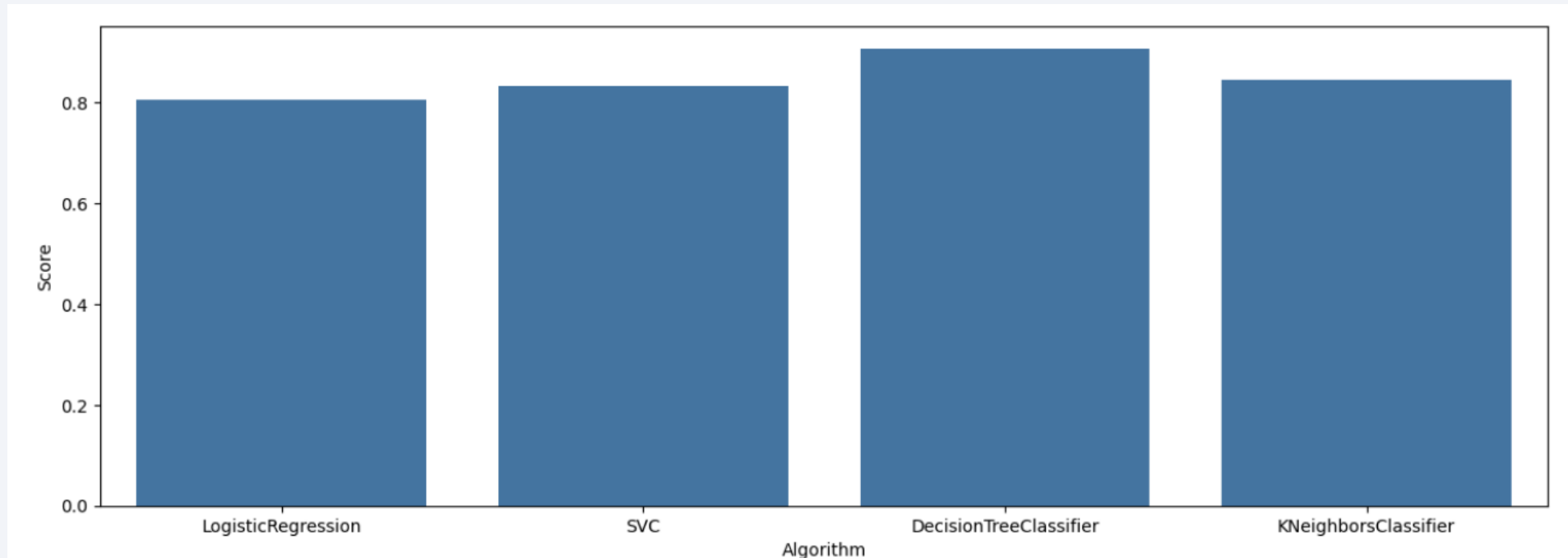
- Payload mass between 0kg and 9600kg v1.1 booster performed the worst
- B4 and B5 boosters had the best success rate in the same payload range.



Section 5

Predictive Analysis (Classification)

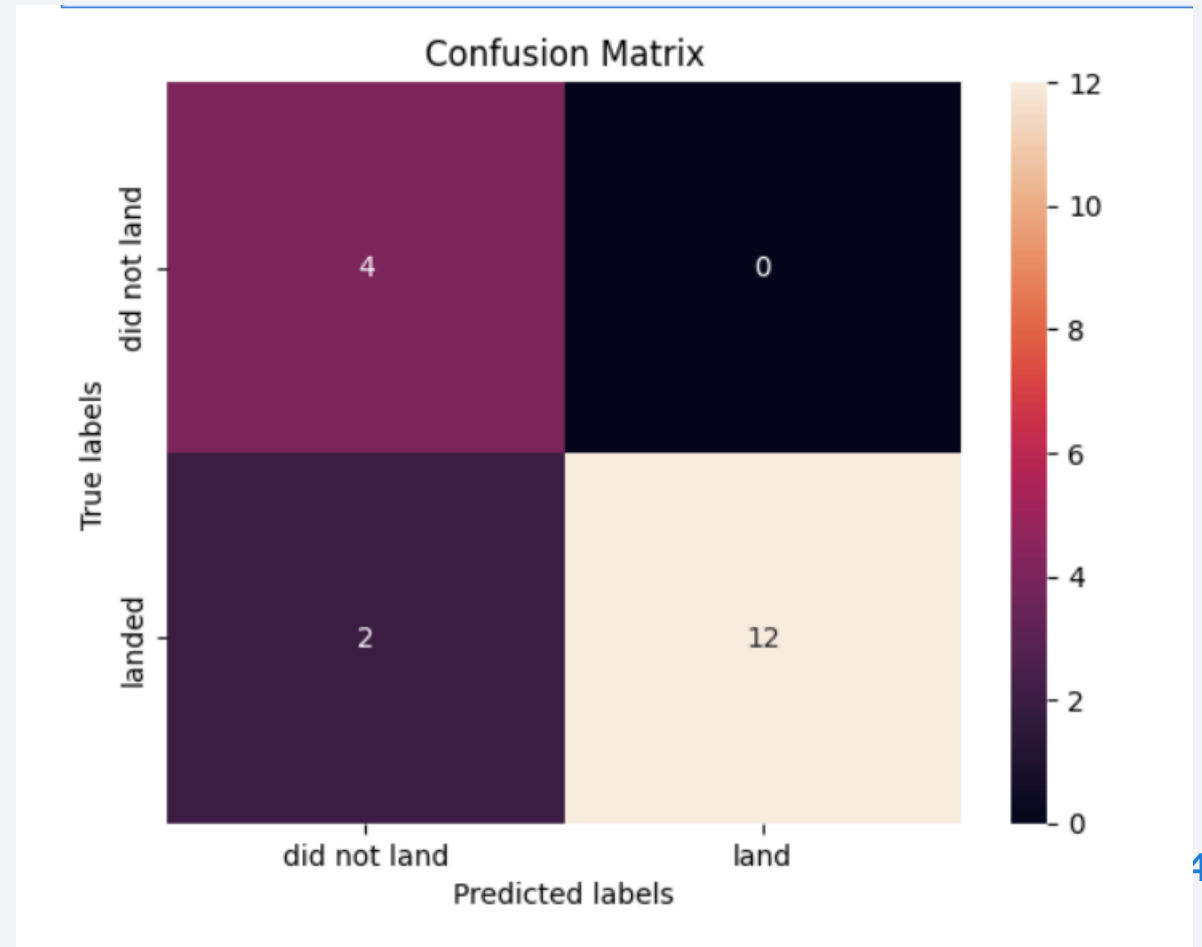
Classification Accuracy



- The model which has the highest classification accuracy of the 4 models used is DecisionTreeClassifier

Confusion Matrix

- The best performing model is the **Decision Tree** using the GridSearchCV best scores since all models have the same accuracy Scores on the test data set.
- Overview:
 - 9 observations predicted as success landing (true positive)
 - 4 observations predicted as failed landing (true negative)
 - 0 observations predicted as success landing (false positive)
 - 2 observations predicted as failed landing (false negative)



Conclusions

- SpaceX project was to see if we can determine if the first stage of Falcon 9 launch will land, we can determine the cost of a launch
- Several Falcon 9 launch features such as the payload mass and orbit type also appears to be a factor; even with more massive payloads, the first stage often returns successfully.
- Several Machine learning algorithm were used on a test set to predict the outcome of the Falcon 9 launch.
- With the help of maps, plots and tables we were able to navigate though the Falcon 9 Launch features and understand the importance on the launch outcome.
- The Decision Tree algorithm performed the best among the 4 machine learning algorithms using the GridSearchCV Best Score value.

Appendix

- **Data Sources:**

- SpaceX :**

- dataset_part_1.csv

- **Wikipedia: List of Falcon 9 and Falcon Heavy launches Wikipage updated on 9th June 2021**
 - **Data after wrangling: dataset_part_2.csv**
 - **Geographical data: spacex_launch_geo.csv**
 - **Interactive data source: spacex_launch_dash.csv**

Thank you!

